

技能图谱应用

- 技能来源: Common core standard
- 技能图谱: [📄 数学课标整理表格 副本](#)
- Stats:
 - 数据来源: Common Core State Standards for Mathematics
 - 覆盖范围:
- 技能: https://github.com/kleeeeeea/learn-claude-code-js/tree/main/custom_skill_space/math-standards
- 应用:
 - 知识点阐释:

代码块

```
1 SKILL_PROMPT='查一下 5 年级 Fractions 领域有哪些课标' bash
  custom_skill_space/create_and_run_skill.sh
2 [skill] 385 条 -> /Users/l/other_git_repos/calculus-
  quest/nodejs_react_lib/learn-claude-code/custom_skill_space/math-
  standards/references/standards.csv
3 [skill] 总览 -> /Users/l/other_git_repos/calculus-
  quest/nodejs_react_lib/learn-claude-code/custom_skill_space/math-
  standards/references/index.md
4 [skill] SKILL.md -> /Users/l/other_git_repos/calculus-
  quest/nodejs_react_lib/learn-claude-code/custom_skill_space/math-
  standards/SKILL.md
5 [skill] skill 目录: /Users/l/other_git_repos/calculus-
  quest/nodejs_react_lib/learn-claude-code/custom_skill_space/math-standards
6      /Users/l/other_git_repos/calculus-quest/nodejs_react_lib/learn-claude-
  code/custom_skill_space/math-standards:
7      references
8      SKILL.md
9
10     /Users/l/other_git_repos/calculus-quest/nodejs_react_lib/learn-claude-
  code/custom_skill_space/math-standards/references:
11     index.md
12     standards.csv
13 [skill] 测试 prompt: 查一下 5 年级 Fractions 领域有哪些课标
14 [skill] pi --skill /Users/l/other_git_repos/calculus-
  quest/nodejs_react_lib/learn-claude-code/custom_skill_space/math-standards -
  p ... (provider/credential 走 piagent_space 那套)
```

15 [piagent] 跳过安装 (PIAGENT_SKIP_INSTALL)

16 [piagent] 配置来源: /Users/l/other_git_repos/calculus-quest/nodejs_react_lib/learn-claude-code/.env

17 [piagent] provider ark-plan -> https://ark.cn-beijing.volces.com/api/plan (glm-5.3) 写入 /Users/l/.pi/agent/models.json

18 [piagent] 启动默认模型 -> ark-plan/glm-5.3 (/Users/l/.pi/agent/settings.json)

19 [piagent] 会话目录 -> /Users/l/other_git_repos/calculus-quest/nodejs_react_lib/learn-claude-code/piagent_space/sessions

20 [piagent] pi=/Users/l/.nvm/versions/node/v24.15.0/bin/pi
node=/Users/l/.nvm/versions/node/v24.15.0/bin/node

21 [piagent] 工作目录: /Users/l/other_git_repos/calculus-quest/nodejs_react_lib/learn-claude-code

22 5 年级 Fractions 领域全称 ****Number and Operations–Fractions**** (编号前缀 5.NF) , 共 ****7 条****课标, 分属两个 cluster。以下为课标原文及 PDF 页码:

23

24 *## Cluster 1: Use equivalent fractions as a strategy to add and subtract fractions (利用等值分数进行分数加减法)*

25

26 ****5.NF.1**** (P36)

27 > Add and subtract fractions with unlike denominators (including mixed numbers) by replacing given fractions with equivalent fractions in such a way as to produce an equivalent sum or difference of fractions with like denominators.

28

29 用等值分数替换异分母分数 (含带分数) , 使其变成同分母后再加减。

30

31 ****5.NF.2**** (P36)

32 > Solve word problems involving addition and subtraction of fractions referring to the same whole, including cases of unlike denominators.

33

34 解决涉及同整体分数加减 (含异分母情形) 的应用题。

35

36 *## Cluster 2: Apply and extend previous understandings of multiplication and division to multiply and divide fractions (拓展乘除法运算认知, 进行分数乘除法)*

37

38 ****5.NF.3**** (P36–37)

39 > Interpret a fraction as division of the numerator by the denominator ($a/b = a \div b$). Solve word problems involving division of whole numbers leading to answers in the form of fractions or mixed numbers.

40

41 把分数解释为分子除以分母 ($a/b = a \div b$) , 并解决结果为分数或带分数的整数除法应用题。

42

43 ****5.NF.4**** (P36–37)

44 > Apply and extend previous understandings of multiplication to multiply a fraction or whole number by a fraction.

45 > a. Interpret the product $(a/b) \times q$ as a parts of a partition of q into b
equal parts; equivalently, as the result of a sequence of operations $a \times q \div b$.

46 > b. Find the area of a rectangle with fractional side lengths by tiling it
with unit squares of the appropriate unit fraction side lengths, and show
that the area is the same as would be found by multiplying the side lengths.

47

48 分数乘法：将 $(a/b) \times q$ 解释为把 q 均分为 b 份取其中 a 份；用单位分数正方形拼铺求分数
边长长方形的面积。

49

50 **5.NF.5** (P36–37)

51 > Interpret multiplication as scaling (resizing), by:

52 > a. Comparing the size of a product to the size of one **factor** on the basis
of the size of the other **factor**, without performing the indicated
multiplication.

53 > b. Explaining why multiplying a given number by a fraction greater than 1
results **in** a product greater than the given number; explaining why
multiplying a given number by a fraction less than 1 results **in** a product
smaller than the given number.

54

55 把乘法理解为缩放：不计算就比较乘积与因数的大小；解释乘以大于/小于 1 的分数对结果的影
响。

56

57 **5.NF.6** (P36–37)

58 > Solve real world problems involving multiplication of fractions and mixed
numbers.

59

60 解决涉及分数和带分数乘法的实际问题。

61

62 **5.NF.7** (P36–37)

63 > Apply and extend previous understandings of division to divide unit
fractions by whole numbers and whole numbers by unit fractions.

64 > a. Interpret division of a unit fraction by a non-zero whole number, and
compute such quotients.

65 > b. Interpret division of a whole number by a unit fraction, and compute
such quotients.

66 > c. Solve real world problems involving division of unit fractions by non-
zero whole numbers and division of whole numbers by unit fractions.

67

68 单位分数除以非零整数、整数除以单位分数，并解决相应实际问题。

69

70 如需查看某条课标在 PDF 中的完整上下文或对齐教学内容，可以告诉我。

71 [skill] 验证 skill 是否真的被加载

72 ok 回答里有 7 个能在 standards.csv 里对上的课标编号

73 [skill] skill 加载成功

◦ 知识点识别

代码块

1